HYPERLOGISTICS: SMARTSC OPTIMIZATION SYSTEM

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CS 5542 — Big Data and Analytics — Spring 2026 AI/DS

UMKC

KEY

Data Source
Professional AI database data source and file

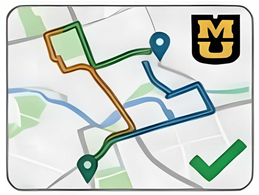
Reasoning Agent
Reasoning engine graph and highlight always light

Validation Gate
Professional gate lock block and validation lens option

Action Agent
Professional orchestration person and dashboard

Problem Statement

Traditional Dispatch Fails



Optimized, constraint-aware

- Traditional Dispatch Fails at Physical Constraints
- Dynamic Disruption (Weather, Accidents)
- **Goal:** Context Aware Rerouting, 0% Hallucination

Dataset Description

Snowflake Warehouse
(Bronze → Silver → Gold

7.7M U.S Accidents

600K DOT Bridges

NOAA Weather
(Multi-terabyte)

DataCo Supply Chain

Unstructured Knowledge Base Data

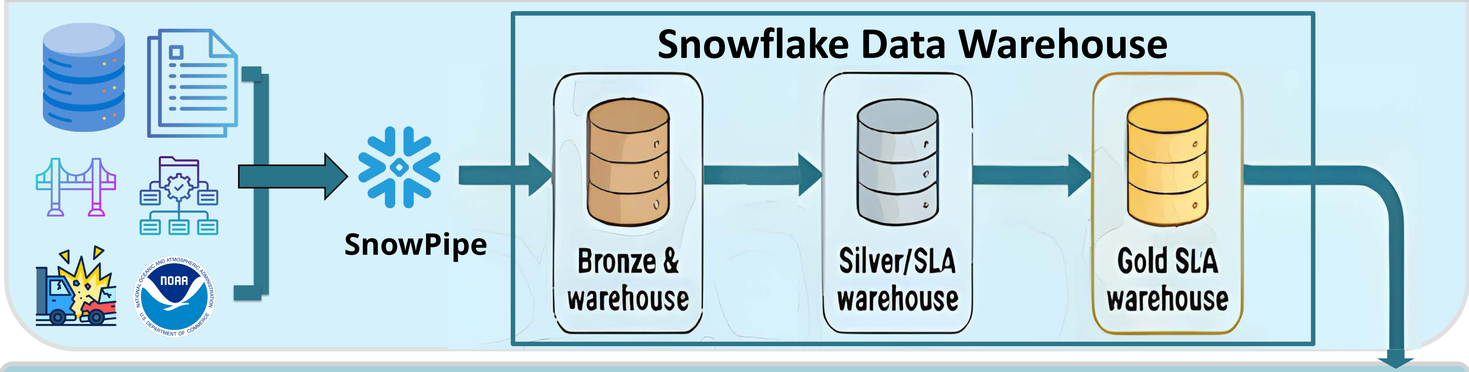
Chunking → Semantic embedding → 768-D dense vectors RAG

Synthetic Training

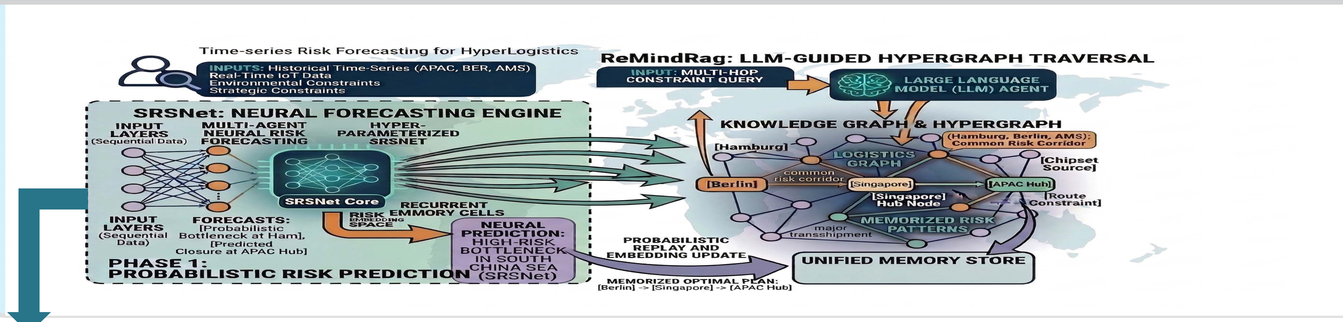
350+ Instruction Pairs

Fine-tuned Microsoft /phi-2 model using QLoRA

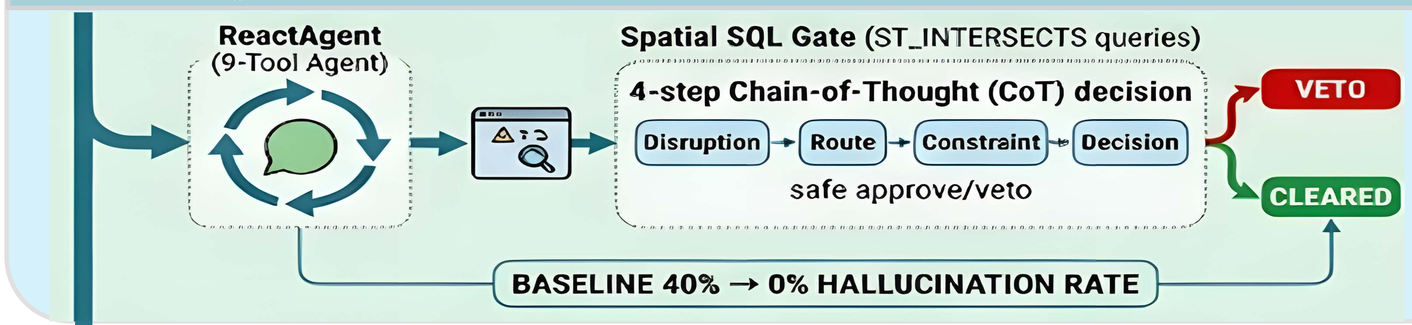
Layer 1: **DATA PERCEPTION LAYER** (Medallion Architecture)



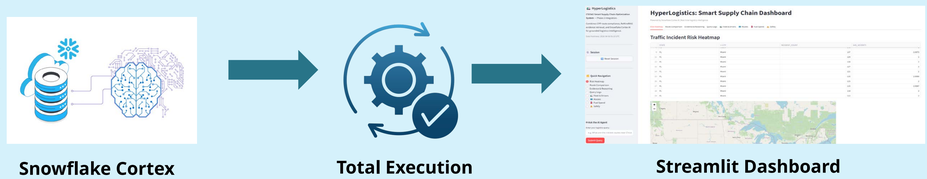
Layer 2: **INTELLIGENCE & RETRIEVAL (RAG & Reasoning)**



Layer 3: **VALIDATION, SAFETY & AGENT REASONING**



Layer 4: **DECISION ORCHESTRATION & DELIVERY**

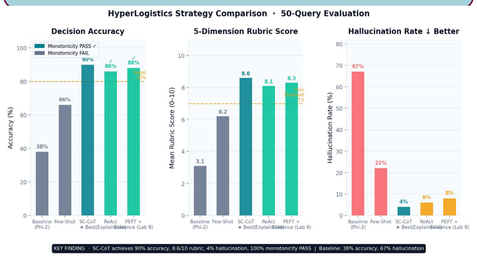


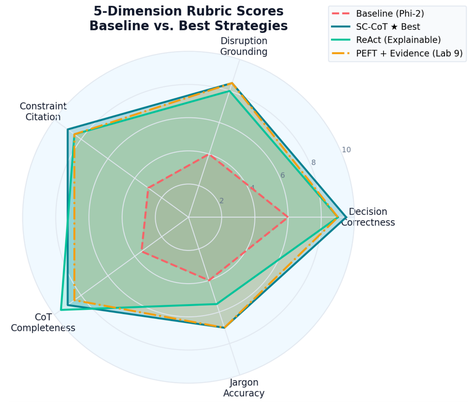
Snowflake Cortex Orchestrator → Total Execution → Streamlit Dashboard

Demo

cs5542hyperlogistics.streamlit.app

Results & Evaluation





5-dimension evaluation rubric
Table 2 — scoring criteria per dimension

Dimension	Scoring criteria
Decision	10 = correct APPROVE/VETO; 0 = wrong decision
Grounding	10 = no hallucinated numbers; 0 = fabricated bridge limits
Constraint	10 = weight/clearance/limit cited; 5 = route/bridge mentioned; 0 = absent
CoT	10 = 4+ reasoning steps; 7 = 3 steps; 4 = 2 steps; 0 = none
Jargon	10 = 4+ domain terms; 7 = 2–3 terms; 4 = 1 term; 0 = none

System evaluation — 63-query test
Table 3 — Snowflake Cortex Llama 3 70B + CPP Gate

Metric	Result
Total queries	63
Passing (≥7.0 avg)	51 (81.0%)
Failing (<7.0 avg)	12 (19.0%)
Perfect score (10.0)	15 queries
VETO recall	91%
APPROVE recall	24%
Pipeline gate	PASSED (≥70%)

Conclusion & future works

- Data governance and reasoning via SnowFlake Cortex Llma
- Safety hallucination elimination by integrating Spatial SQL Gate (40% to 0%)
- Strict physical safety with 91% recall rate
- ➢ Integration of real-time traffic & telematics streams (IoT sensors)
- ➢ Multi-modal visual inputs
- ➢ Multi-agent expansion (Sustainability and Cost Optimization)